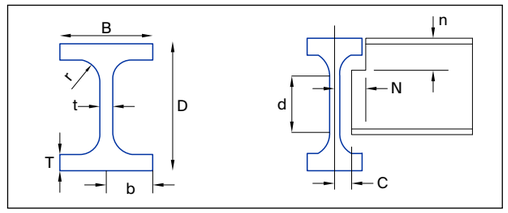
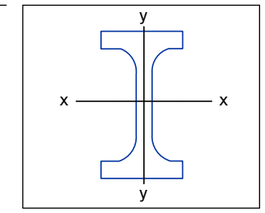
2.2
Wide Flange Beams (HE)



Section designation	Dimensions												
	Mass per Metre	Depth of Section	Width of Section	Thickness		Root	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
				Flange	Web				Flange	Web	End clearance	Notch	
		D	B	T	t	r	d	A	b/T	d/t	C	N	n
	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
HE 300 B	117	300.0	300.0	19.0	11.0	27.0	208.0	149	18.9	6.18	8	156	46
HE 300 C	177	320.0	305.0	29.0	16.0	27.0	208.0	225	13.0	4.05	10	156	56
HE 300 M	238	340.0	310.0	39.0	21.0	27.0	208.0	303	9.90	3.01	13	156	66
HE 320 AA	74.2	301.0	300.0	11.0	8.0	27.0	225.0	94.6	28.1	10.82	6	156	38
HE 320 A	97.6	310.0	300.0	15.5	9.0	27.0	225.0	124	25.0	7.65	7	156	44
HE 320 B	127	320.0	300.0	20.5	11.5	27.0	225.0	161	19.6	5.72	8	156	48
HE 320 C	186	340.0	305.0	30.5	16.0	27.0	225.0	237	14.1	3.85	10	156	58
HE 320 M	245	359.0	309.0	40.0	21.0	27.0	225.0	312	10.7	2.93	13	154	68
HE 340 AA	78.9	320.0	300.0	11.5	8.5	27.0	243.0	100	28.6	10.33	6	156	40
HE 340 A	105	330.0	300.0	16.5	9.5	27.0	243.0	134	25.6	7.17	7	156	44
HE 340 B	134	340.0	300.0	21.5	12.0	27.0	243.0	171	20.3	5.44	8	154	50
HE 340 M	248	377.0	309.0	40.0	21.0	27.0	243.0	316	11.6	2.93	13	154	68
HE 360 AA	83.7	339.0	300.0	12.0	9.0	27.0	261.0	107	29.0	9.88	7	156	40
HE 360 A	112	350.0	300.0	17.5	10.0	27.0	261.0	143	26.1	6.74	7	156	46
HE 360 B	142	360.0	300.0	22.5	12.5	27.0	261.0	181	20.9	5.19	8	154	50
HE 360 M	250	395.0	308.0	40.0	21.0	27.0	261.0	319	12.4	2.91	13	154	68
HE 400 AA	92.4	378.0	300.0	13.0	9.5	27.0	298.0	118	31.4	9.10	7	156	40
HE 400 A	125	390.0	300.0	19.0	11.0	27.0	298.0	159	27.1	6.18	8	156	46
HE 400 B	155	400.0	300.0	24.0	13.5	27.0	298.0	198	22.1	4.84	9	154	52
HE 400 M	256	432.0	307.0	40.0	21.0	27.0	298.0	326	14.2	2.90	13	154	68
HE 450 AA	99.7	425.0	300.0	13.5	10.0	27.0	344.0	127	34.4	8.74	7	156	42
HE 450 A	140	440.0	300.0	21.0	11.5	27.0	344.0	178	29.9	5.58	8	156	48
HE 450 B	171	450.0	300.0	26.0	14.0	27.0	344.0	218	24.6	4.46	9	154	54
HE 450 M	263	478.0	307.0	40.0	21.0	27.0	344.0	335	16.4	2.90	13	154	68
HE 500 AA	107	472.0	300.0	14.0	10.5	27.0	390.0	137	37.1	8.41	7	156	42
HE 500 A	155	490.0	300.0	23.0	12.0	27.0	390.0	198	32.5	5.09	8	154	50

Dimensions	EN10365
Specification	EN10025
Size Range	HE 100 AA to HE 1000 x 584



	Properties											
Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
1.73	25,200	8,560	13.0	7.58	1,680	571	1,870	870	7.706	14.1	1.69	189
1.78	41,000	13,700	13.5	7.81	2,560	901	2,930	1,370	7.835	10.0	2.90	604
1.83	59,200	19,400	14.0	8.00	3,480	1,250	4,080	1,910	7.947	7.88	4.39	1,410
1.74	16,400	4,960	13.2	7.24	1,090	331	1,200	506	7.617	21.8	1.04	53.6
1.76	22,900	6,980	13.6	7.49	1,480	466	1,630	710	7.761	17.5	1.51	112
1.77	30,800	9,240	13.8	7.57	1,930	616	2,150	939	7.813	14.2	2.07	230
1.82	48,700	14,400	14.3	7.81	2,860	947	3,270	1,440	7.945	10.3	3.45	688
1.87	68,100	19,700	14.8	7.95	3,800	1,280	4,440	1,950	8.028	8.20	5.00	1,510
1.78	19,600	5,180	14.0	7.18	1,220	346	1,340	529	7.675	22.6	1.23	60.0
1.79	27,700	7,440	14.4	7.46	1,680	496	1,850	756	7.837	17.9	1.82	131
1.81	36,700	9,690	14.7	7.53	2,160	646	2,410	986	7.882	14.5	2.45	263
1.90	76,400	19,700	15.6	7.90	4,050	1,280	4,720	1,950	8.085	8.70	5.58	1,510
1.81	23,000	5,410	14.7	7.12	1,360	361	1,500	553	7.710	23.3	1.44	67.1
1.83	33,100	7,890	15.2	7.43	1,890	526	2,090	802	7.893	18.1	2.18	153
1.85	43,200	10,100	15.5	7.49	2,400	676	2,680	1,030	7.935	14.8	2.88	298
1.93	84,900	19,500	16.3	7.83	4,300	1,270	4,990	1,940	8.115	9.21	6.14	1,510
1.89	31,200	5,860	16.3	7.06	1,650	391	1,820	600	7.778	24.8	1.95	81.3
1.91	45,100	8,560	16.8	7.34	2,310	571	2,560	873	7.951	19.0	2.94	193
1.93	57,700	10,800	17.1	7.40	2,880	721	3,230	1,100	7.987	15.7	3.82	361
2.00	104,000	19,300	17.9	7.70	4,820	1,260	5,570	1,930	8.155	10.3	7.41	1,520
1.98	41,900	6,090	18.2	6.92	1,970	406	2,180	624	7.780	27.4	2.57	91.4
2.01	63,700	9,460	18.9	7.29	2,900	631	3,220	966	8.016	20.0	4.15	250
2.03	79,900	11,700	19.1	7.33	3,550	781	3,980	1,200	8.039	16.7	5.26	448
2.10	132,000	19,300	19.8	7.59	5,500	1,260	6,330	1,940	8.176	11.6	9.25	1,530
2.08	54,600	6,310	20.0	6.79	2,320	421	2,580	649	7.756	29.9	3.30	103
2.11	87,000	10,400	21.0	7.24	3,550	691	3,950	1,060	8.050	20.8	5.64	318